ORACLE TO AZURE DATABASE FOR POSTGRESQL

Migration assessment

Contoso Store

A preliminary, evidence-backed disposition: which Azure PostgreSQL target to assess first, how to convert and move the schema, what the converter will not do, and how much of the work the source code says is there.

1,523
schema objects scanned

18,209
lines of PL/SQL

133–265
person-days by hand

95–227
with AI-assisted conversion

5 and 10 min per unit. The AI figure is a model and needs the converter pilot.

Status	Preliminary. Provisional until an architect and an assessment tool validate it.
Evaluated	2026-10-12
Versions	Knowledge base v1.0 · rules v1.0 · sources checked 2026-10-12
Produced by	oracle-to-postgres-migration-advisor, built on the design of fredgis/sql-migration-advisor

1 Executive summary

Azure Database for PostgreSQL flexible server on PostgreSQL 17 is the target to assess first. Status **provisional**, confidence **medium**. It is a preliminary disposition, not a verdict.

1,523	790	733	133–265	95–227
objects	**clean**	**need human attention**	**person-days by hand**	**with AI-assisted conversion**
25,498 lines of code	52% of objects	453 partial, 279 review, 1 blocker	5 and 10 min per unit. The AI figure is a model and needs the converter pilot.	

What the scan found

- **1,523 objects and 25,498 lines of code**, of which 18,209 are PL/SQL, across 28 script files. 90 package bodies hold 474 routines.
- **80 objects carry package-level session state, 3 use autonomous transactions, and 1 touch the file system.** None has a direct PostgreSQL equivalent.
- **43 objects are created by dynamic DDL** and 10 statements are never terminated in the source (a stray `;` inside a comment). A scan that reads only top-level `CREATE` statements would have missed the first and mis-split the second.
- **1 construct has no supported equivalent on flexible server:** Reference / system / virtual-column partitioning.

What the advisor recommends

- **Target:** Azure Database for PostgreSQL flexible server, PostgreSQL 17. Rule `OP-RANK`: Lowest score among the surviving targets (10.25).
- **Alternative:** none named. No other family survives the rules.
- **Schema and code:** `MS_VSCODE_AI`. The Microsoft converter does not list Oracle 23ai or 26ai, so a pilot on 10 to 15 objects decides it.
- **Effort:** 133 to 265 person-days by hand; about 95 to 227 if the AI-assisted converter drafts the code (a model, section 9).
- **Data:** `ADF_COPY`, offline. **No Microsoft service moves Oracle rows to PostgreSQL by change capture.** The converter does not move a row either.

The five things most likely to hurt

1. **Associative arrays (INDEX BY)** (130 objects, 190 uses): PL/pgSQL has no sparse or string-keyed array.
2. **CONNECT BY pseudo-columns, NOCYCLE, ORDER SIBLINGS** (39 objects, 157 uses): Oracle detects cycles with NOCYCLE. PostgreSQL recurses until it runs out of memory unless the query guards itself.
3. **Package-level session state** (80 objects, 123 uses): PostgreSQL has no package state. Values that persisted for a session either disappear or move into settings.
4. **SAVE EXCEPTIONS / SQL%BULK_EXCEPTIONS** (10 objects, 41 uses): Oracle continues past failing rows. PostgreSQL aborts the statement, so a quarantine pattern becomes a slower loop.
5. **MEMBER / STATIC / MAP / ORDER methods** (17 objects, 27 uses): Standalone functions taking the composite as first argument

> **What this report is not.** It reads source text only: no row counts, workload, performance or application. Effort covers reading, rewriting and a basic test per routine. Data movement, tuning, application changes and cutover are separate workstreams (section 9).

2 Recommendation

PRELIMINARY RECOMMENDATION

OP-RANK

Azure Database for PostgreSQL flexible server

PostgreSQL 17 · status **provisional** · confidence **medium**

SCHEMA AND CODE	DATA MOVEMENT	CUTOVER DOWNTIME	ALTERNATIVE
`MS_VSCODE_AI` Microsoft converter	`ADF_COPY` offline	**minutes to a couple of hours** + validation	**none named** no other family survives

Phase A: all eight target families

TARGET	STATE	RULE	WHY
Azure Database for PostgreSQL flexible server	◐ eligible with remediation	`OP-FEATURE-MATRIX`	8 listed Oracle feature(s) are carried by a workaround, a redesign, or in a way the vendor does not document: redesign (3), workaround (5).
Flexible server elastic cluster	✕ unsupported	`OP-DR`	An elastic cluster offers a single read replica and no documented cross-region path.
Azure HorizonDB for PostgreSQL	✕ unsupported	`OP-AVAILABILITY`	HorizonDB publishes no SLA percentage.
PostgreSQL on Azure virtual machines (self-managed)	⊘ excluded by preference	`OP-MANAGEMENT`	The customer prefers a fully managed service.
PostgreSQL on AKS (CloudNativePG)	⊘ excluded by preference	`OP-MANAGEMENT`	The customer prefers a fully managed service.
Azure Cosmos DB for PostgreSQL	✕ unsupported	`OP-COSMOS`	New deployments are blocked and the service retires 2029-03-31. Existing customers move to elastic clusters.
Oracle AI Database@Azure	✓ eligible	none	Nothing consumed rules it out.
Oracle Database on Azure virtual machines	⊘ excluded by preference	`OP-MANAGEMENT`	The customer prefers a fully managed service.

The state that wins is the most severe one any rule produced. **Unsupported** is a technical incompatibility. **Excluded by preference** names an answer the customer can change. Rule IDs are defined in `reference/decision-rules.md`.

What the chosen target needs from the application

ORACLE FEATURE IN THE SCHEMA	CARRIED AS	HOW
DBMS_SCHEDULER chains, windows, job classes, event-based jobs	redesign	pg_cron runs a schedule and nothing more. Chains and events move to Azure Automation, Logic Apps or Functions
Object types with UNDER and member methods	redesign	Composite types carry no methods and no substitutability. Flatten with a discriminator

ORACLE FEATURE IN THE SCHEMA	CARRIED AS	HOW
Fast-refresh or ON COMMIT materialized views	redesign	Core PostgreSQL refreshes completely. pg_ivm is on the allow-list and may cover simple views; test each
UTL_FILE and DIRECTORY objects	workaround	azure_storage reads and writes blobs. There is no server file system to write to
Virtual Private Database (DBMS_RLS)	workaround	Row level security with CREATE POLICY. Policy functions become USING expressions
PRAGMA AUTONOMOUS_TRANSACTION	workaround	dblink self-connection, or move the write outside the transaction. The converter emits the dblink form
Package-level variables that live for the session	workaround	Session settings (set_config), or session_variable, which blocks in-place major version upgrade
Interval, reference, composite or global-index partitioning	workaround	Declarative partitioning with pg_partman for interval. Reference partitioning and global indexes have no equivalent, so the key is redesigned

Largest risk. 3 Oracle feature(s) have no equivalent on this target and need an application or data-model change: DBMS_SCHEDULER chains, windows, job classes, event-based jobs; Object types with UNDER and member methods; Fast-refresh or ON COMMIT materialized views.

3 Estate profile

The interview answers the recommendation rests on. **Stated** answers are assumptions written for this example from what the lab's documentation says; a real engagement replaces them. **Scan** values come from the source code and win over a stated value. PL/SQL volume bands are under 5,000, under 50,000, under 250,000 and above lines of code; this schema has 18,209. A field never answered is `UNKNOWN` and is never read as "none".

FIELD	VALUE	FROM	FIELD	VALUE	FROM
Scope	`SINGLE_DB`	stated	Customer-managed keys	`CMK_NO`	stated
Source location	`ON_PREM`	stated	Largest database (planned)	`UNDER_1_TB`	stated
Oracle version	`ORA_23AI_26AI`	stated	Peak IOPS	`UNDER_20K_IOPS`	stated
Topology	`SINGLE_INSTANCE`	stated	Peak compute	`UP_TO_96_VCORES`	stated
Edition	`NOT_SURE`	unknown	Availability target	`HIGH_99_95 (99.95% (same-zone HA))`	stated
Character set	`AL32UTF8`	stated	Cross-region recovery	`ASYNC_CROSS_REGION`	stated
Intent	`MIGRATE_NOW`	stated	Feature dependencies (stated)	`LIST_FEATURES`	scan
Main driver	`ORACLE_LICENSE_COST`	stated	PL/SQL volume	`MODERATE`	scan
Management model	`MANAGED_PAAS`	stated	Oracle app tier	`NONE_CONFIRMED`	stated
Application changeability	`APP_LIMITED`	stated	Cutover downtime	`MINIMAL`	stated
ISV supports PostgreSQL	`UNKNOWN`	unknown	Bandwidth	`GOOD_BANDWIDTH`	stated
Compliance	`EU_DATA_BOUNDARY`	stated	Private connectivity	`PRIVATE_LINK_READY`	stated
Azure region	`swedencentral`	stated	Converter prerequisites	`FOUNDRY_AND_COPILOT_READY`	stated
Preview services	`PREVIEW_REFUSED`	stated			

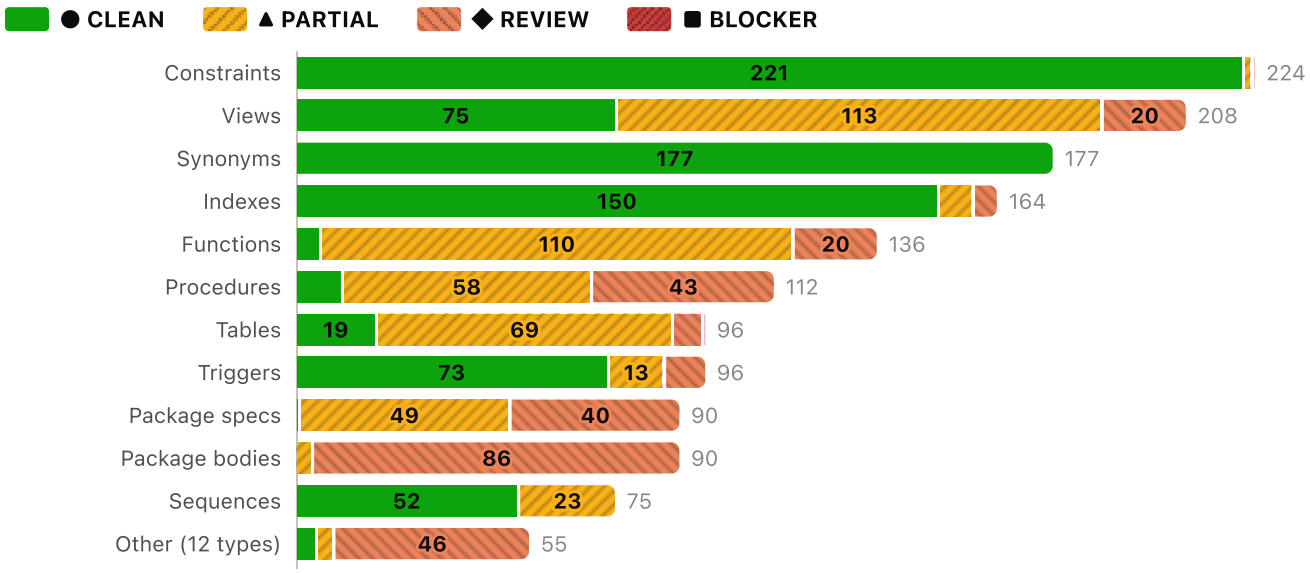
FIELD	VALUES	FROM
Oracle features found	`FILE_ACCESS_FROM_DATABASE, VPD, SCHEDULER_ADVANCED, AUTONOMOUS_TRANSACTIONS, PACKAGE_STATE, OBJECT_TYPE_INHERITANCE, MV_FAST_REFRESH, HEAVY_PARTITIONING`	scan
Evidence in hand	`SCHEMA_SCAN, VSCODE_DISCOVERY`	scan

No decision-driving unknown remains. 2 fields are unanswered (Edition, ISV supports PostgreSQL); under the current rules none of them changes the target. Confidence is capped at medium because the skill reads no artefact from the estate.

4 Inventory and complexity

1,523 objects in 28 script files. Each object is scored against 256 catalogued Oracle constructs and put in the worst class any of them gives it.

Objects by type, split by class



Clean is mechanical. **Partial** converts but can drift in an edge case. **Review** needs a design decision or a hand rewrite on any PostgreSQL target. **Blocker** has no supported equivalent on flexible server. Class is a property of constructs, not of size. Types with fewer than 20 objects are folded into Other in the chart only. Units are the effort score of section 9.

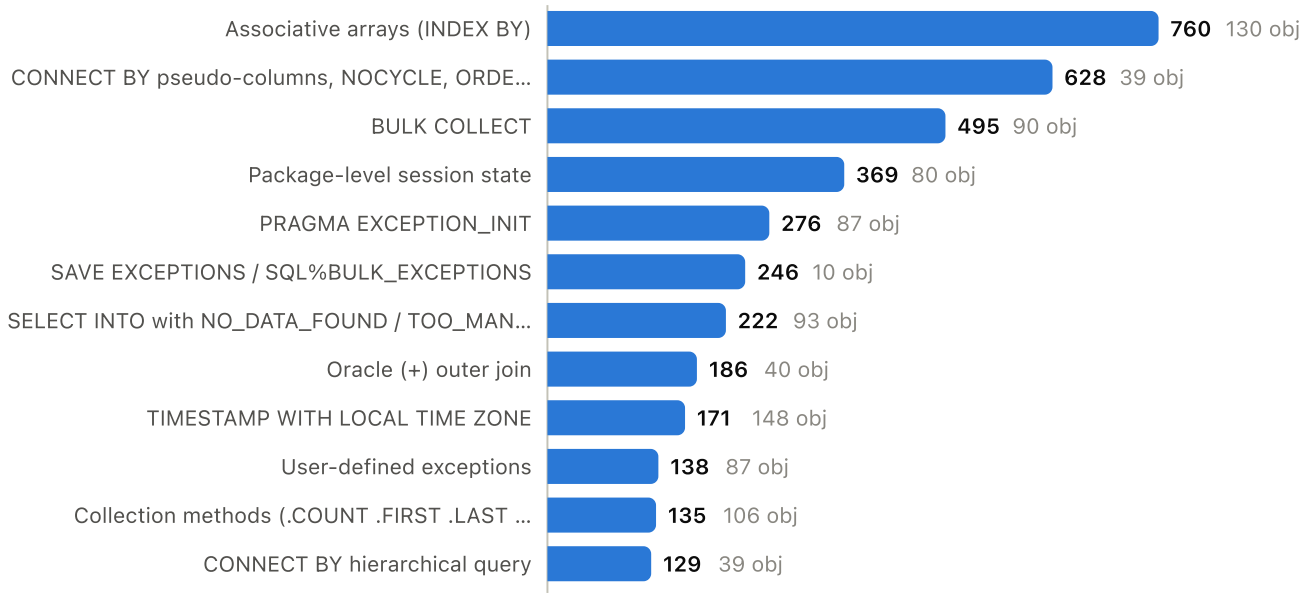
All types, with the numbers behind the chart

Type	Objects	Units	Clean	Part.	Rev.	Block.
Constraints	224	32	221	2	1	0
Views	208	916	75	113	20	0
Synonyms	177	18	177	0	0	0
Indexes	164	82	150	8	6	0
Functions	136	1,009	6	110	20	0
Procedures	112	1,361	11	58	43	0
Tables	96	471	19	69	7	1
Triggers	96	679	73	13	10	0
Package specs	90	772	1	49	40	0
Package bodies	90	5,008	0	4	86	0
Sequences	75	30	52	23	0	0
Types	18	179	4	2	12	0

Type	Objects	Units	Clean	Part.	Rev.	Block.
Type bodies	8	188	0	0	8	0
Scheduler jobs	8	53	0	0	8	0
Materialized views	6	140	0	1	5	0
MV logs	3	9	0	0	3	0
Schedules	3	20	0	0	3	0
VPD policies	3	112	0	0	3	0
Programs	2	10	0	0	2	0
Users	1	13	0	1	0	0
Refresh groups	1	5	0	0	1	0
Job classes	1	1	1	0	0	0
Chains	1	6	0	0	1	0
Total	**1,523**	**11,113**	**790**	**453**	**279**	**1**

5 What drives the effort

Constructs by estimated effort (units)



One unit is five minutes of a PostgreSQL expert (Ora2Pg's convention); 84 units are a person-day. Weights are Ora2Pg 25.0's own where it scores a construct, and marked *advisor* in the catalog where it does not.

Where the constructs sit: objects affected, by category and class

● CLEAN ▲ PARTIAL ◆ REVIEW ■ BLOCKER

	● CLEAN	▲ PARTIAL	◆ REVIEW	■ BLOCKER
PL/SQL language	14	1,106	286	·
SQL dialect	28	172	62	·
Data types and semantics	4	276	21	·
semantics	·	83	·	·
Security	·	85	6	·
Storage and DDL	·	51	20	1
Supplied packages	34	2	18	·
Integration	·	·	1	·

Silent semantic traps present in this schema

These compile, run, and return different answers. A converter that reports "converted" has not seen them.

CONSTRUCT	OBJECTS	WHY IT BITES
Package-level session state (H-43)	80	PostgreSQL has no package state. Values that persisted for a session either disappear or move into settings.
SAVE EXCEPTIONS / SQL%BULK_EXCEPTIONS (H-10)	10	Oracle continues past failing rows. PostgreSQL aborts the statement, so a quarantine pattern becomes a slower loop.

CONSTRUCT	OBJECTS	WHY IT BITES
SELECT INTO with NO_DATA_FOUND / TOO_MANY_ROWS	93	Without STRICT, PostgreSQL neither raises NO_DATA_FOUND nor TOO_MANY_ROWS. The handler silently never runs.
COMMIT / ROLLBACK / SAVEPOINT inside routines	90	A function that commits has to become a procedure, and so do its callers.
SYSDATE / SYSTIMESTAMP	358	now() is the transaction start time. SYSDATE is the statement time.
Materialized view fast refresh / ON COMMIT (H-15)	4	Freshness guarantees change. ON COMMIT views are stale until the next refresh.
ROWID / UROWID	13	Code that updates or deletes by ROWID needs a real key.
ROWNUM (H-30)	62	ROWNUM is assigned before ORDER BY. A naive LIMIT conversion returns a different set of rows.

6 Where to start: hotspots

The 12 heaviest package bodies

Package bodies are where the work and the risk concentrate: 90 of them hold 5,008 of 11,113 units (45%). The Microsoft converter converted 27 of 90 in the lab's measured run, the weakest of any object type.

PACKAGE BODY	ROUTINES	LINES	STATE VARS	UNITS	CLASS
PKG_CATALOG	11	161	·	119.0	◆ REVIEW
PKG_ETL_EXPORT	3	107	·	115.4	◆ REVIEW
PKG_UTILS	15	114	2	97.5	◆ REVIEW
PKG_GEN_IFC_OUT_012	8	100	1	87.2	◆ REVIEW
PKG_GEN_IFC_OUT_024	8	100	1	87.2	◆ REVIEW
PKG_GEN_IFC_OUT_036	8	100	1	87.2	◆ REVIEW
PKG_REPORTING	6	125	·	80.9	◆ REVIEW
PKG_GEN_IFC_OUT_004	6	104	1	80.9	◆ REVIEW
PKG_GEN_IFC_OUT_016	6	104	1	80.9	◆ REVIEW
PKG_GEN_IFC_OUT_028	6	104	1	80.9	◆ REVIEW
PKG_GEN_IFC_OUT_008	5	88	1	72.8	◆ REVIEW
PKG_GEN_IFC_OUT_020	5	88	1	72.8	◆ REVIEW

The 12 heaviest other objects

OBJECT	TYPE	LINES	UNITS	MAIN CONSTRUCTS
MV_SALES_DAILY_STORE	Materialized view	53	53.0	Materialized view fast refresh / ON COMMIT ×5; DBMS_MVIEW / DBMS_REFRESH; DBMS_OUTPUT ×9
VPD_CUSTOMER_PII_MASK	VPD policy	11	48.0	Virtual Private Database (DBMS_RLS) ×3; DBMS_RLS call ×3
T_MONEY	Type body	52	43.6	MEMBER / STATIC / MAP / ORDER methods ×4; EXECUTE IMMEDIATE; SELECT INTO with NO_DATA_FOUND / TOO_MANY_ROWS

OBJECT	TYPE	LINES	UNITS	MAIN CONSTRUCTS
FN_CATEGORY_PATH	Function	33	42.3	CONNECT BY pseudo-columns, NOCYCLE, ORDER SIBLINGS ×6; CONNECT BY hierarchical query ×3; SELECT INTO with NO_DATA_FOUND / TOO_MANY_ROWS ×2
INVENTORY_MOVEMENT	Table	23	37.4	Composite partitioning ×7; Interval partitioning; INTERVAL YEAR TO MONTH / DAY TO SECOND
T_PARTY	Type	6	32.1	Object type inheritance (UNDER / NOT FINAL) ×3; Object types (AS OBJECT); MEMBER / STATIC / MAP / ORDER methods
V_CATEGORY_TREE	View	18	32.0	CONNECT BY pseudo-columns, NOCYCLE, ORDER SIBLINGS ×7; CONNECT BY hierarchical query; String concatenation (||) with NULL
VPD_SALES_ORDER_BY_STORE	VPD policy	10	32.0	Virtual Private Database (DBMS_RLS) ×2; DBMS_RLS call ×2
VPD_CUSTOMER_BY_COUNTRY	VPD policy	10	32.0	Virtual Private Database (DBMS_RLS) ×2; DBMS_RLS call ×2
T_POSTAL_ADDRESS	Type body	20	31.7	MEMBER / STATIC / MAP / ORDER methods ×2; Empty string literal ('' is NULL in Oracle) ×5; Overloaded package and type routines
MV_SUPPLIER_PERFORMANCE	Materialized view	54	31.6	Materialized view fast refresh / ON COMMIT ×3; DBMS_OUTPUT ×5; SYSDATE / SYSTIMESTAMP ×6
V_EMPLOYEE_REPORTING_LINE	View	18	28.0	CONNECT BY pseudo-columns, NOCYCLE, ORDER SIBLINGS ×6; CONNECT BY hierarchical query

7 Target configuration

Flexible server has no superuser and no custom extensions. Each extension is allow-listed through the `azure.extensions` parameter, then created per database; those that load a library must also be in `shared_preload_libraries`, which restarts the server. Do this **before** the first conversion run: if `plpgsql_check` is missing the converter silently skips deep validation.

EXTENSION	STATUS	PRELOAD	OBJECTS	NEEDED BY
orafce	✓ on the allow-list	no	754	TYPE-SYSDATE, TYPE-VARCHAR2-NVARCHAR2-BYTE-CHAR, SQL-NVL +5
hstore	✓ on the allow-list	no	130	PLSQL-ASSOC-ARRAY
session_variable	✓ on the allow-list	no	80	PLSQL-PACKAGE-STATE
pg_cron	✓ on the allow-list	yes, restart	18	DDL-MV-FAST-REFRESH, PKG-DBMS-SCHEDULER
pg_hint_plan	✓ on the allow-list	yes, restart	13	SQL-HINT
pg_partman	✓ on the allow-list	yes, restart	3	DDL-PARTITION-INTERVAL
dblink	✓ on the allow-list	no	3	PLSQL-AUTONOMOUS-TXN
tablefunc	✓ on the allow-list	no	1	SQL-PIVOT
azure_storage	✓ on the allow-list	yes, restart	1	PKG-UTL-FILE

Settings and choices that cannot be changed later

- **PostgreSQL 17 or 18. PostgreSQL 18 lacks some extensions, so check the allow-list first. Never PostgreSQL 14 or older.** The converter needs a scratch database on PostgreSQL 15 or later.
- **High availability:** target 99.95% (same-zone HA). Zone-redundant gives 99.99%, same-zone 99.95%, doubling compute and storage cost. swedencentral is not in the blocked or unavailable lists, which are volatile; confirm in the portal.
- **Network mode** (private access or public with private endpoint) is fixed at creation. So are customer-managed keys and geo-redundant backup.
- **Collation and character set:** source is AL32UTF8; create the database as UTF8. Use collation `C` on a comparison copy or ordering differs from Oracle.
- **No tablespaces.** Oracle tablespace placement is flattened. 0 objects carry physical storage clauses that are dropped.

> **Sizing is not done here.** Compute and storage depend on AWR or ASH data this tool does not read. Collect a representative month of AWR before choosing a SKU. The Learn module and the Microsoft converter size nothing either.

8 Outside the converter

The Microsoft converter converts schema and PL/SQL. It does **not** move rows, and it does not convert the objects below. Each one the scan found is a manual workstream with an owner.

NOT CONVERTED BY THE MICROSOFT TOOL	FOUND IN THIS SCHEMA	WHAT TO DO INSTEAD
Scheduler jobs, chains and programs	**15**	Rebuild on pg_cron; chains and events need an orchestrator (Logic Apps, Functions, Automation).
VPD policies (DBMS_RLS)	**3**	Recreate as row level security policies and set the session context they read.
Application contexts	**35**	Replace with a custom setting namespace. The guarantee that only a trusted package writes it has to be rebuilt.
Materialized view logs	**3**	Drop. They exist only to support fast refresh.
Bitmap, global-partitioned and domain indexes	**4**	Redesign the access path: btree, GIN or BRIN, local indexes.
Users, roles and privileges	**1**	Script roles and grants by hand; move authentication to Microsoft Entra ID.
Database links	none found	oracle_fdw or postgres_fdw, one foreign table per remote object.
Advanced Queuing	none found	Azure Service Bus, or a queue table read with SKIP LOCKED.
External tables	none found	azure_storage reads blobs. file_fdw is not on the allow-list.
System-event triggers	none found	login_hook for logon logic, event triggers for DDL.
Java and external C procedures	none found	Move the logic to the application tier.

Source: Schema conversion limitations. The converter also never converts system schemas (SYS, SYSTEM, XDB, MDSYS, CTXSYS, WMSYS), and "Converted" does not guarantee the code compiles, is complete, or behaves the same.

Oracle components around the database

Forms, Reports, APEX, OBIEE, ODI, WebLogic and mod_plsql do not migrate with the database. They are application work. The profile confirms there are none.

9 Effort and risk

11,113
cost units
5 minutes each

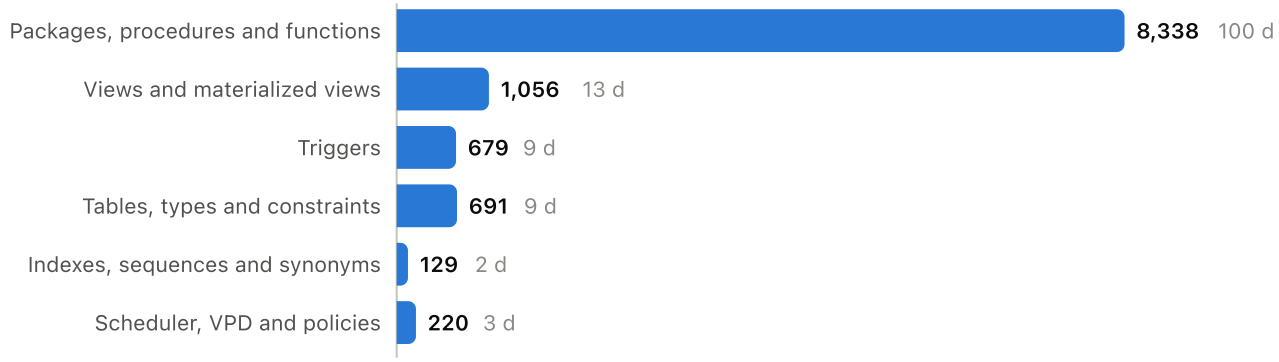
133
person-days at 5 min/unit
an expert, per Ora2Pg

265
person-days at 10 min/unit
Ora2Pg's suggestion for a first migration

C-5
Ora2Pg level
C: above the 5-day limit

The figure is Ora2Pg's convention reimplemented, not a quote. Each routine costs 2 units to test, 1 unit per 1,000 characters to read, and the sum of its construct scores. A cost of 11,113 units is about 133 person-days at an expert's pace and 265 at the pace Ora2Pg suggests for a first migration. **The range is the two paces, not a statistical interval.** AI-assisted conversion shifts the mix between machine and human time, and the lab recorded the machine time (2 h 56 min for 1,185 objects) but not the human review time.

Units by workstream



If the AI-assisted converter does the first pass

The Microsoft converter (PostgreSQL extension for VS Code, a Foundry `gpt-5.2` deployment, a Copilot seat) writes the first version of each object, and people review it. The figure below counts the same cost units after that change. It is **a model with stated assumptions, not a measurement**: no vendor publishes a productivity figure to anchor it, and the real number comes from the pilot.

133–265
person-days by hand
the figure above

95–227
with the AI-assisted converter
converter run measured on this schema

71%–86%
of the manual effort remains
testing, decisions, review

26
person-day floor
what the converter cannot touch

WHERE THE UNITS GO	UNITS	WHAT THE MODEL ASSUMES
Reading and rewriting the converter can draft	6,447	A person reviews the draft at 25% to 50% of the manual cost. An object the converter does not convert costs the full manual amount.

WHERE THE UNITS GO	UNITS	WHAT THE MODEL ASSUMES
Constructs that need a design decision	2,764	The converter proposes, a person decides: 2 times the review cost, never above the manual cost.
The testing allowance	1,662	Not reduced. Converted is not equivalent, and the same model that converted an object cannot be the only check on it.
Blockers and objects the converter does not read	239	Not reduced: jobs, VPD policies, users, and constructs with no supported equivalent.
Setup, pilot and conversion runs	3 to 5 days	A fixed allowance: Foundry quota, Copilot seats, the scratch server, the pilot and the batch runs.

Objects not converted: 32% measured on the lab's converter run, by object type. The low end uses the 25% review cost and 5 minutes a unit, the high end 50% and 10 minutes, the same two paces as the manual figure. The converter's own failures in the lab run were mostly timeouts and lock contention in the scratch database, so a healthy run may convert more than this.

> **The converter does not list this Oracle release (23ai and 26ai).** The figure above holds only if the 10 to 15 object pilot shows the converter works on it. If it does not, the manual figure stands.

Where the AI-assisted figure falls, by workstream (person-days)

WORKSTREAM	BY HAND	WITH AI-ASSISTED CONVERSION
Packages, procedures and functions	99.3 to 198.5	70.4 to 166.9
Views and materialized views	12.6 to 25.1	9.7 to 22.6
Triggers	8.1 to 16.2	5.2 to 13.0
Tables, types and constraints	8.2 to 16.4	3.1 to 11.1
Indexes, sequences and synonyms	1.5 to 3.1	0.7 to 2.4
Scheduler, VPD and policies	2.6 to 5.2	2.6 to 5.2

How sensitive it is

Person-days, low to high, for other review costs and conversion rates. The pilot replaces these guesses with numbers for this estate.

REVIEW COST	0% NOT CONVERTED	32% NOT CONVERTED	50% NOT CONVERTED
15%	48 to 94	76 to 150	92 to 182
25%	62 to 122	85 to 169	99 to 196
50%	97 to 193	110 to 218	117 to 232
75%	117 to 232	123 to 244	126 to 251

> **What the AI does not change.** Every converted object is still read and tested. The design decisions (scheduler chains, object types with inheritance, fast-refresh views) stay with people. Data movement, application changes, performance tuning, cutover and rollback are outside both figures. The Foundry tokens and Copilot seats are real costs, and this report states no cost.

Not estimated by a source scan

WORKSTREAM	WHY A SCAN CANNOT SIZE IT	WHAT WOULD
Data movement and rehearsals	Row counts, LOB sizes and change rate are not in the DDL.	Table sizes from the data dictionary; a timed rehearsal load.
Performance tuning	Plans and workload are not in the source. Hints are ignored by PostgreSQL.	AWR or ASH and a representative query set.
Application changes	Callers of refcursors, error codes and package state live outside the database.	An application dependency review; the preview app-conversion tool as a first pass.
Testing at scale	Differential testing needs data and queries.	The ten most important differential questions run on both sides.
Cutover and fallback	Depends on downtime tolerance. Microsoft offers no change-capture service, so a short window is a gap.	A rehearsed cutover; a retained, read-only Oracle copy as the way back.

Largest risk

3 Oracle feature(s) have no equivalent on this target and need an application or data-model change: DBMS_SCHEDULER chains, windows, job classes, event-based jobs; Object types with UNDER and member methods; Fast-refresh or ON COMMIT materialized views.

10 Data movement

> **No Microsoft migration service moves Oracle rows to PostgreSQL.** Azure Database Migration Service has no Oracle to PostgreSQL scenario, the PostgreSQL migration service accepts PostgreSQL sources only, and the Microsoft converter converts schema and code only. The routes below move rows offline. Microsoft offers no change-capture service for Oracle, so this advisor names no product for it.

Downtime tolerance **MINIMAL** with a database under 1 TiB and good bandwidth allows an offline bulk load with a Microsoft route. If the window shrinks until it needs change capture, that is a gap in what Microsoft offers: the account team can say what options exist. Whatever is chosen then needs ARCHIVELOG mode with supplemental logging on the source, which the plan lists.

ROUTE	ROLE	STATUS	NOTES
`ADF_COPY`	transport	✓ available	Data Factory Oracle connector v2.0. Moves rows. No DDL and no change capture.
`ORACLE_FDW_COPY`	transport	✓ available	INSERT SELECT over oracle_fdw. Needs a network path from the PostgreSQL server to Oracle.

What the schema says the load must handle

SIGNAL	TABLES	WHY IT MATTERS
Tables in the schema	96	Parallelism is by table, partition and key range.
Partitioned tables	4	Partitions load in parallel; the target partition keys must include the primary key.
CLOB or BLOB columns	36	Large values dominate load time. Test the largest LOB table first and size the batches by the largest value.

SIGNAL	TABLES	WHY IT MATTERS
LONG columns	1	LONG cannot be read by most change-capture products and needs a special initial load.
XMLType columns	2	Stored as xml; check the content parses on load.
Global temporary tables	3	Hold no rows to move; the definition must exist in every session.
Index-organized tables	3	Load in key order into a heap table, then CLUSTER.

11 Validation and prerequisites

"Converted" is not "equivalent". Microsoft's own documentation says converted code must be independently validated, and the same model that converted it can also review it. Plan validation as its own workstream; the steps are in section 12.

CHECK	WHAT IT CATCHES	NOTES
plpgsql_check	Static errors in PL/pgSQL, bad references, type mismatches	Allow-list and preload it, restart, and keep it for production. Absent on HorizonDB.
Differential testing	A converted query that is wrong and quiet	The same queries on both sides, result sets diffed. Start with ROWNUM, (+) joins and empty-string tests.
Row comparison	Rows that were truncated, rounded or re-encoded in the copy	oracle_fdw foreign tables, or hashed row queries run on both sides. Needs primary keys. Collation C or ordering differs.
Plan comparison	Queries that were fast through a hint or Exadata offload	Hints are comments to PostgreSQL. pg_hint_plan is available.

12 Prerequisite plan

37 steps follow from the recommendation and the scan: 5 before committing to a path, 22 before the first conversion run, 6 before the cutover rehearsal, 2 at cutover, 2 after the move. The plan is generated from `reference/prerequisite-catalog.json`, so it changes when the recommendation does. Application steps come from the features the chosen target carries by workaround or redesign. Owners are suggestions.

Before committing to a path

Cheap, mostly paperwork. Each answer can change the recommendation.

AREA	STEP	OWNER
Oracle	**Export the DDL and PL/SQL source for every schema in scope** The schema scan reads source text and needs no database connection. Use SQLcl DDL output or DBMS_METADATA.GET_DDL, one file per object type, with grants and synonyms. Judgement: the scan accepts any script that holds the DDL, so the export tool is the customer's choice	Oracle DBA
Oracle	**Collect a representative month of AWR or ASH data** Compute, IOPS, throughput and connection counts come from here. This advisor does not size a target, and no Microsoft converter does either. Judgement: sizing is outside the advisor by design	Oracle DBA

AREA	STEP	OWNER
Commercial	**Read the Oracle contract for support end date, notice period and any Oracle-on-Azure terms** The saving from leaving Oracle depends on the contract, and Oracle's licensing on Azure is a contract question, not an architecture one. Keeping Oracle on Azure virtual machines is the benchmark any saving is measured against. Sources: Oracle licensing for authorized cloud environments; Oracle Lifetime Support: Oracle Database technology chart	Licence management
Commercial	**Ask about delivery funding before the project is scoped** Frontier Accelerate (formerly Azure Accelerate), Cloud Accelerate Factory and FastTrack have different eligibility, and Microsoft's pages disagree on the programme name. Quote both names and confirm with the programme owner. Nomination and approval are required. Sources: Azure Accelerate; Cloud Accelerate Factory; FastTrack for Azure FAQ; Partner Center announcements, May 2026	Account team
Conversion tooling	**Pilot the Microsoft converter on 10 to 15 representative objects first** Microsoft lists Oracle 12.1 to 21c and does not list 23ai or 26ai. Include a package with state, a view with CONNECT BY, a trigger, and one object you expect to fail. Reruns of a completed conversion are not supported. Sources: Schema conversion overview; Schema conversion limitations	Migration engineer

Before the first conversion run

Long lead times live here: quota, seats, networking, extensions.

AREA	STEP	OWNER
Oracle	**Create a read account for the converter** CONNECT, SELECT_CATALOG_ROLE, SELECT ANY DICTIONARY, SELECT on SYS.ARGUMENT$ and SYS.V_$RESOURCE_LIMIT, and the sessions parameter above 10. It needs dictionary access, not table data. The thin driver reaches Oracle 12.1 and later; native encryption or checksums switch it to thick mode (Instant Client). Sources: Schema conversion tutorial and prerequisites; Schema conversion overview	Oracle DBA
Azure	**Deploy gpt-5.2 in Microsoft Foundry with more than 1,000,000 tokens per minute of quota** Quota requests are the long pole, so file them first. Authenticate by API key or Microsoft Entra ID (Foundry User role). A private endpoint is recommended. Run one project at a time, and for a large schema run VS Code on an Azure VM near Foundry and the scratch server. Sources: Schema conversion tutorial and prerequisites; Oracle migration in the PostgreSQL extension for VS Code	Cloud platform team
Azure	**Assign a GitHub Copilot seat (Pro+, Business or Enterprise) to each converter user** The tutorial lists it as a requirement and the extension's own prerequisite page does not, so treat it as required for the Ask and Fix steps. Sources: Schema conversion tutorial and prerequisites; Oracle migration in the PostgreSQL extension for VS Code	Developer platform
Azure	**Create a scratch flexible server on PostgreSQL 15 or later** Match the major version to the production target. HorizonDB is not supported as the scratch database. The tool runs DROP SCHEMA ... CASCADE on matching scratch schemas, so never point it at a database that holds anything else. Deploy converted code to a separate non-production database for testing. Sources: Schema conversion tutorial and prerequisites; Schema conversion limitations	Cloud platform team

AREA	STEP	OWNER
Azure	**Allow-list the extensions the schema needs, then preload the ones that load a library** Set azure.extensions, then create each extension per database. plpgsql_check, pg_cron and pg_hint_plan must also be in shared_preload_libraries, which restarts the server. Do this before the first conversion run: with plpgsql_check missing the converter skips its deep validation and still reports Converted. Set a database-level search_path for the orafce schemas. Sources: Extension versions by PostgreSQL major; Extension considerations; plpgsql_check; orafce	Cloud platform team
Azure	**Settle the choices that cannot be changed after the server exists** Network mode (private access or public with private endpoint), customer-managed keys, geo-redundant backup and the major version. High availability mode is easier to change but doubles compute and storage cost. Sources: Private networking; Data encryption; Backup and restore	Architect
Azure	**Choose storage as capacity, IOPS and throughput that can only grow** Premium SSD v2 sets the three separately. Storage cannot shrink. Size from the AWR evidence, with headroom for the load, not from the Oracle allocation. Sources: Storage; Premium SSD v2; Azure Database for PostgreSQL limits	Cloud platform team
Network and security	**Open a private path from the converter workstation and the loader to both databases** Use a bastion or a private endpoint rather than public addresses. The loader needs the Oracle listener port and the PostgreSQL port, and any VM that runs the converter should sit near Foundry and the scratch server. Sources: Private networking	Network team
Network and security	**Script roles, grants and password policy by hand and move sign-in to Microsoft Entra ID** The Microsoft converter does not convert users, roles, profiles or privileges. Flexible server has no superuser and a restricted administrator role. Sources: Schema conversion limitations; Access control and privileges	Security
Data movement	**Set up the Data Factory Oracle connector v2.0** Connector v1.0 was removed on 2026-03-31, so create linked services with v2.0. Check whether the Oracle listener is reachable from the Azure integration runtime or needs a self-hosted one. The connector moves rows only: no DDL and no change capture. Sources: Azure Data Factory Oracle connector; ADF connector release stages	Migration engineer
Data movement	**Tune the load for large objects** LOB columns dominate load time. Test the largest LOB table first, size the load batches by the largest value, and expect BLOB to land as bytea. Judgement: load behaviour depends on the route and is not published for the Microsoft routes	Migration engineer
Data movement	**Give LONG columns a special initial load** Change-capture products typically cannot read LONG. Plan a bulk load for the table and a way to catch rows that change during it. Judgement: depends on the capture product, which Microsoft does not offer	Migration engineer
Data movement	**Redesign keys on partitioned tables so the primary key includes the partition key** PostgreSQL has no global indexes: a unique or primary key on a partitioned table must contain the partition key. Interval partitioning needs pg_partman. Reference partitioning has no equivalent. Sources: Extension versions by PostgreSQL major; Schema conversion limitations	Data architect

AREA	STEP	OWNER
Data movement	**Handle index-organised tables, XMLType columns and global temporary tables explicitly** Load index-organised tables in key order into a heap table, then CLUSTER. Check that XMLType content parses as xml on load. Global temporary tables hold no rows to move, but the definition must exist for every session. Judgement: derived from the object kinds the scan found	Migration engineer
Application	**DBMS_SCHEDULER chains, windows, job classes, event-based jobs: redesign** pg_cron runs a schedule and nothing more. Chains and events move to Azure Automation, Logic Apps or Functions Sources: Extension versions by PostgreSQL major	Application owner
Application	**Object types with UNDER and member methods: redesign** Composite types carry no methods and no substitutability. Flatten with a discriminator Sources: Schema conversion overview	Application owner
Application	**Fast-refresh or ON COMMIT materialized views: redesign** Core PostgreSQL refreshes completely. pg_ivm is on the allow-list and may cover simple views; test each Sources: Extension versions by PostgreSQL major	Application owner
Application	**UTL_FILE and DIRECTORY objects: workaround** azure_storage reads and writes blobs. There is no server file system to write to Sources: Extension versions by PostgreSQL major	Application owner
Application	**Virtual Private Database (DBMS_RLS): workaround** Row level security with CREATE POLICY. Policy functions become USING expressions Sources: Schema conversion limitations	Application owner
Application	**PRAGMA AUTONOMOUS_TRANSACTION: workaround** dblink self-connection, or move the write outside the transaction. The converter emits the dblink form Sources: Extension versions by PostgreSQL major; Schema conversion overview	Application owner
Application	**Package-level variables that live for the session: workaround** Session settings (set_config), or session_variable, which blocks in-place major version upgrade Sources: Extension versions by PostgreSQL major; Major version upgrade	Application owner
Application	**Interval, reference, composite or global-index partitioning: workaround** Declarative partitioning with pg_partman for interval. Reference partitioning and global indexes have no equivalent, so the key is redesigned Sources: Extension versions by PostgreSQL major	Application owner

Before the cutover rehearsal

Prove the load, the validation and the rollback on real volume.

AREA	STEP	OWNER
Data movement	**Rehearse the full load on production-sized data and time it** Row counts, LOB sizes and change rate are not in the DDL, so the window cannot be estimated from the schema. Rehearse the full path, including sequence resets and constraint creation. Judgement: the scan cannot see data volume	Migration engineer

AREA	STEP	OWNER
Validation	**Run plpgsql_check across every converted routine** It finds static errors, bad references and type mismatches without running the code. It works in production and is not available on HorizonDB. Converted does not mean the check ran if the extension is missing. Sources: plpgsql_check; Extension versions by PostgreSQL major	Migration engineer
Validation	**Run the same queries on Oracle and PostgreSQL and diff the result sets** This is the only route that catches a converted query that is wrong and quiet. Start with ROWNUM, (+) joins, empty-string tests, string concatenation with NULL, DECODE with NULL and DATE values with a time part. The same model that converted the code can also review it, so validate independently. Sources: Schema conversion overview; Schema conversion best practices	QA lead
Validation	**Compare rows with oracle_fdw foreign tables or hashed row queries** Needs a primary or unique key on each table. Create the comparison copy with collation C, or ordering differs from Oracle. Sources: Extension versions by PostgreSQL major	QA lead
Validation	**Compare plans and timings for the queries that carry the business** Oracle hints are comments to PostgreSQL. pg_hint_plan is available where a hint is the right fix. Queries that were fast through Exadata offload need indexes, partitioning and tuning instead. Sources: Extension versions by PostgreSQL major	Performance engineer
Application	**Test connection handling against the target's connection limits** The default maximum connections depend on the vCores of the SKU, and Oracle shared-server and pooling behaviour does not carry over. Use the built-in connection pooler or an application pool. Sources: Azure Database for PostgreSQL limits	Application owner

At cutover

The window, the handoff point and the way back.

AREA	STEP	OWNER
Data movement	**Fix the handoff point and reset every sequence** Record the SCN for a consistent snapshot. Reset sequences to at least the source's last value. Judgement: the SCN and the sequence reset are properties of the load, not of one route	Migration engineer
Operations	**Keep Oracle read-only and decide the way back before the window opens** No source read for this project describes a supported reverse replication from PostgreSQL to Oracle. If the fallback needs one, name the tool, prove it in the rehearsal and put its licence in the plan. Judgement: absence of a documented reverse path	Architect

After the move

Operate the target as PostgreSQL, not as Oracle.

AREA	STEP	OWNER
Operations	**Set backup retention and test a point-in-time restore** A point-in-time restore creates a new server. It is not Flashback and there is no AS OF query. Geo-redundant backup is a creation-time choice. Sources: Backup and restore	Cloud platform team

AREA	STEP	OWNER
Operations	**Put the PostgreSQL major version lifecycle in the run book** Standard support ends on a published date for each major, and in-place major version upgrade is blocked while some extensions, such as session_variable, are installed. Sources: Version policy; Major version upgrade; Extended support	Cloud platform team

13 Cost levers, programmes and next steps

Cost levers

- Retiring Oracle licences and annual support: the saving depends on the customer's Oracle contract. Confirm with Oracle contract and licence management.
- Azure Hybrid Benefit does not apply to Azure Database for PostgreSQL, so there is no licence to bring.
- Reserved capacity (one or three years) covers compute only. Microsoft pages quote different maxima, so use the pricing calculator for the region and SKU.
- High availability doubles compute and storage cost. Use zone-redundant HA only where the availability target needs it.
- Burstable plus stop and start suits dev and test only. It is not for production and has no HA.
- Storage can only grow, never shrink. Premium SSD v2 sets capacity, IOPS and throughput separately.
- Keeping Oracle on Azure virtual machines is the benchmark: two Azure vCPUs are one Oracle processor licence with multithreading on, with no core factor discount. Confirm with Oracle.

No cost figure appears here on purpose. A figure needs measured sizing and stated pricing assumptions, and this scan has neither. Use the Azure pricing calculator after sizing.

Microsoft programmes

Quote both names: Microsoft is renaming the programme, and its pages disagree.

- Frontier Accelerate for Azure and for Databases (formerly Azure Accelerate): partner-nominated delivery funding and credits, advertised at 50% of year-one usage, subject to application and approval. A May 2026 Partner Center notice names Oracle to Azure PostgreSQL. Engage through the account team or a specialised partner.
- Cloud Accelerate Factory: zero-cost Microsoft delivery assistance alongside a partner. Oracle to PostgreSQL is not named. Nomination is required and the value is not guaranteed.
- FastTrack for Azure: free remote guidance for nominated projects above $5,000 a month. Database scope is not stated.

Next actions

1. Run the discovery step of the Microsoft converter for the object inventory and invalid objects.
2. Pilot 10 to 15 representative objects end to end before committing to an estimate.
3. Allow-list the extensions the schema needs through azure.extensions, and preload plpgsql_check and pg_cron, before the first conversion run.

Never recommended

ITEM	USE INSTEAD
Azure Data Studio and all its extensions	the PostgreSQL extension for VS Code

ITEM	USE INSTEAD
Azure Data Studio Oracle assessment (DMAO) and schema conversion toolkit (DSCT)	the VS Code conversion tool
Azure Database for PostgreSQL Single Server	flexible server
Azure Arc-enabled PostgreSQL	flexible server
Azure Cosmos DB for PostgreSQL, for new deployments	flexible server elastic clusters
Azure Database Migration Service for Oracle to PostgreSQL	none: no Microsoft migration service moves Oracle rows to PostgreSQL. The offline routes are the Data Factory copy and oracle_fdw
SSMA for Oracle as a route to PostgreSQL	SSMA targets SQL Server and Azure SQL only
Azure Migrate as an Oracle database assessor	the VS Code discovery step, the schema scan
Azure Data Factory Oracle connector v1.0	Oracle connector v2.0
PostgreSQL 11, 12 and 13 as a new target	PostgreSQL 16 to 18
PostgreSQL 14 as a new target	PostgreSQL 16 to 18

A Appendix A. The scan against the lab's declared hard cases

The Contoso Store lab declares which objects carry each of its 43 hard cases. That is a ground truth the scanner did not write. For every declared object, the check is whether the scan flagged a construct mapped to the same case.

92%
object recall
149 of 162 declared objects

31/40
cases fully covered
every declared object found

28/43
class agrees with lab's prediction
14 more cautious, 1 less

What the misses are

- **Recursive PL/SQL that is not CONNECT BY** (`fn_manager_chain`, `fn_location_depth`). The lab lists them under H-06 as the alternative to the construct. They contain none.
- **Indirect uses**: a scheduler job that only calls a routine that uses UTL_FILE; the empty-string-is-NULL cases (H-38) where the code tests `IS NULL` on a column whose type is declared in another file.
- **The lab's own documentation ahead of its code**: two objects named under a case whose source does not contain the construct (comments only).

This measures recall of the declared locations, not precision. Precision was checked separately by reading matched lines for every high-weight construct.

Where this catalog disagrees with the lab's own predictions

An independent research pass checked the lab's predictions against vendor documentation and the lab's measured conversion run. It agreed on 84 constructs, extended 11, and disagreed on nine.

CASE	LAB SAYS	THIS CATALOG SAYS	EVIDENCE
T–03 byte-sized lengths	multibyte names overflow after conversion	clean	PostgreSQL `varchar(n)` counts characters, so limits loosen. The real risks are index row size and downstream Oracle consumers.
T–14 `||` and NULL	both databases return NULL	partial, high impact	Oracle treats NULL as empty in concatenation: `'a' || NULL` is `'a'`. PostgreSQL returns NULL. Silent.
T–13 NULL sort order	a difference	clean	Both put NULLs last ascending and first descending. No work.
H–08 KEEP / NTH_VALUE FROM LAST	clean	partial	The lab's own run left `v_product_price_rank` not converted with a best-effort flag. Plain analytics stay clean.
H–31 DECODE	clean	partial	Oracle treats two NULLs as equal in DECODE. The lab documents three such sites itself, and two views were not converted.
H–10 SAVE EXCEPTIONS	partial	review	Per-row failure handling is a behaviour decision. `pkg_receiving` stayed not converted in the lab run.
H–15 fast-refresh views	no analogue	review, partial analogue	`pg_ivm` is on the Azure allow-list. It does not help here because the base tables are partitioned, so the lab's conclusion stands.

CASE	LAB SAYS	THIS CATALOG SAYS	EVIDENCE
H–19 primary key without the partition key	partial	review	The lab calls it the most disruptive item. It forces a key redesign across foreign keys.
H–34 DBMS_LOB	clean	partial	orafce does not ship DBMS_LOB. The lab's search path names a schema that does not exist.

Where a claim could not be verified (for example whether orafce's `decode` treats NULL as equal) the research marked it unverified, and the catalog does not rely on it.

CASE	HARD CASE	LAB PREDICTED	SCAN CLASS		OBJECTS FOUND
H–01	Packages with overloaded procedures	partial	▲ PARTIAL	✓	3/3
H–02	PRAGMA AUTONOMOUS_TRANSACTION	review task	◆ REVIEW	✓	3/3
H–03	Object types with member methods (and inheritance)	review task	◆ REVIEW	✓	10/10
H–04	VARRAY collections	clean	▲ PARTIAL	≠	4/4
H–05	Nested table collections	review task	◆ REVIEW	✓	4/4
H–06	CONNECT BY hierarchical queries	partial	◆ REVIEW	≠	7/9
H–07	MERGE statements	clean	◆ REVIEW	≠	3/3
H–08	Analytic and window functions	clean	▲ PARTIAL	≠	4/4
H–09	REF CURSOR returns	partial	▲ PARTIAL	✓	2/2
H–10	BULK COLLECT and FORALL	partial	◆ REVIEW	≠	5/5
H–11	Native dynamic SQL (EXECUTE IMMEDIATE)	partial	◆ REVIEW	≠	3/4
H–12	DBMS_OUTPUT	clean	● CLEAN	✓	9/9
H–13	UTL_FILE	review task	◆ REVIEW	✓	1/2
H–14	DBMS_SCHEDULER jobs	review task	◆ REVIEW	✓	no object resolved
H–15	Materialised views with fast refresh and a refresh group	review task	◆ REVIEW	✓	1/1
H–16	Function-based indexes	partial	▲ PARTIAL	✓	2/2
H–17	Virtual columns	partial	▲ PARTIAL	✓	8/8
H–18	Index-organized tables	clean	▲ PARTIAL	≠	3/3
H–19	Range-partitioned tables	partial	◆ REVIEW	≠	3/3
H–20	List-partitioned tables	clean	▲ PARTIAL	≠	2/2
H–21	Global temporary tables	partial	▲ PARTIAL	✓	3/3
H–22	Sequences with NOCACHE and CYCLE	clean	▲ PARTIAL	≠	3/3
H–23	DETERMINISTIC functions	partial	▲ PARTIAL	✓	5/5
H–24	RESULT_CACHE functions	review task	◆ REVIEW	✓	2/2
H–25	Triggers of every timing	partial	▲ PARTIAL	✓	no object resolved
H–26	Compound triggers	review task	◆ REVIEW	✓	3/3
H–27	INSTEAD OF triggers	clean	● CLEAN	✓	3/3
H–28	User-defined exceptions and RAISE_APPLICATION_ERROR	partial	▲ PARTIAL	✓	3/3
H–29	PRAGMA EXCEPTION_INIT	partial	▲ PARTIAL	✓	1/1

CASE	HARD CASE	LAB PREDICTED	SCAN CLASS		OBJECTS FOUND
H–30	ROWNUM versus ROW_NUMBER()	partial	▲ PARTIAL	✓	2/3
H–31	NVL, NVL2, DECODE	clean	▲ PARTIAL	≠	4/5
H–32	Oracle outer-join (+) syntax	partial	▲ PARTIAL	✓	2/2
H–33	LONG columns	partial	▲ PARTIAL	✓	1/1
H–34	CLOB and BLOB columns	clean	▲ PARTIAL	≠	5/5
H–35	XMLTYPE columns	partial	◆ REVIEW	≠	3/3
H–36	INTERVAL types	partial	▲ PARTIAL	✓	7/9
H–37	TIMESTAMP WITH LOCAL TIME ZONE	partial	◆ REVIEW	≠	8/9
H–38	Empty string is NULL	review task	◆ REVIEW	✓	3/6
H–39	SYS_CONTEXT and application contexts	partial	▲ PARTIAL	✓	3/3
H–40	Virtual Private Database policies	review task	◆ REVIEW	✓	7/8
H–41	The synonym layer	partial	● CLEAN	≠	no object resolved
H–42	Pipelined table functions	partial	▲ PARTIAL	✓	2/2
H–43	Package-level session state	review task	◆ REVIEW	✓	2/2

B Appendix B. Does the score predict what the converter did?

The lab ran the Microsoft converter on this schema (2026-09-04, Oracle 23.26, extension 1.30.1, gpt-5.2) and published a per-object result. This joins each scored object to its outcome (**1,167** objects matched by name) and asks whether higher scores went with more failures. The measure is AUC: the chance that a randomly chosen failed object outscored a randomly chosen converted one. 0.5 is a coin flip.

0.41
overall AUC
programmable objects

261
failures that are dependency or best-effort flags
of 373 not converted

The score does not predict the converter's failures, and it should not be read as if it did. In this run, failures follow dependency cascades (an object waits on a package that did not convert) and best-effort flags more than they follow construct difficulty. The lab's own analysis attributes 95% of stated failure reasons to timeouts and lock contention in the scratch database. The cost units are an estimate of *human* reading and rewriting time. They were not designed to predict a tool's success, and this appendix shows they do not.

Outcome by scan class

Class	Objects	Not converted	Rate
● CLEAN	564	173	31%
▲ PARTIAL	399	120	30%
◆ REVIEW	203	79	39%
■ BLOCKER	1	1	100%

Ranking power by object type

Type	Objects	Failed	AUC units	AUC lines
Views	208	143	0.42	0.59
Synonyms	174	90	0.50	0.52
Indexes	158	18	0.53	0.59
Functions	136	6	0.69	0.42
Procedures	112	22	0.44	0.82
Triggers	96	26	0.37	0.34
Tables	95	4	0.81	0.59
Package bodies	90	61	0.51	0.41
Materialized views	6	3	0.78	0.44

AUC above 0.5 means larger or costlier objects failed more. Package bodies are the lab's weakest type (27 of 90 converted) for a reason size does explain: one failing member fails the whole body. The result also says the lab's 79.92% is *a lower bound*, as the lab states.

C Appendix C. Method, scoring and limits

How the scan works

1. **Mask.** Comments, string interiors and SQL*Plus commands are blanked while offsets and line breaks are kept, so a hit is never found in a comment. Only unambiguous SQL*Plus forms are removed: an earlier draft deleted `START WITH`, `CONNECT BY`, `EXIT WHEN` and `DESC` lines.
2. **Split.** Plain SQL ends at `;`. PL/SQL units end at a `/` line. A statement that reaches the next `CREATE` unterminated is recovered and flagged. Objects built by `EXECUTE IMMEDIATE` in installer blocks are found and marked dynamic.

3. **Detect.** 256 catalogued constructs run as patterns or as computed metrics (package state, overloads, string null tests, columns typed with VARRAY or object types).
4. **Score.** Ora2Pg 25.0's model: routine = 1 + 2 test + 1 per 1,000 characters + constructs; view = 1 + constructs. 54 catalog weights are Ora2Pg's own; 202 are this project's, and marked.
5. **Level.** A to C and 1 to 5 as Ora2Pg defines them. The person-day limit that turns B into C is 5 in Ora2Pg's command-line help and 10 in its configuration text; the scan uses 5. At this size the level is C under either limit.
6. **Class.** The worst class of any construct with a weight. Notes that are true of nearly every object (for example `||` and NULL) are counted but do not move an object out of Clean.

Limits

- Static analysis reads source text only. It does not connect to Oracle, so it cannot see row counts, data volumes, runtime behaviour, workload, or objects that exist only in the data dictionary.
- SQL built at run time from data (for example concatenated identifiers inside EXECUTE IMMEDIATE) is invisible. Dynamic DDL in installer blocks is found; dynamic DML in routines is not analysed.
- The cost units are Ora2Pg's convention reimplemented here. They estimate reading and rewriting time. They do not include data movement, performance tuning, application changes, testing at scale, or cutover.
- A construct is counted where its text appears. It is not proof that the construct runs, and absence of a construct is not proof the code is safe.
- Effort bands are a heuristic range, not a quote. Calibrate them against a pilot of 10 to 15 objects converted and reviewed by the team that will do the work.

Script hygiene found in the source

- **10 statements are unterminated** (for example `SEQ_GEN_AUDIT_001`): a `;` typed inside a trailing `--` comment is ignored by Oracle clients, so the statement runs into the next one. The scan recovered the objects; the script should be fixed.
- **43 objects are created by dynamic DDL**, invisible to a scan of top-level statements.

D Appendix D. Sources

Every limit, date and status in this report was read on **2026-10-12** from the 71 pages below. Pages with a volatile answer (regions where zone-redundant HA is blocked, programme names and funding percentages) say so where they are used. Confirm before promising any of them.

Microsoft (58)

Azure Database for PostgreSQL limits
https://learn.microsoft.com/en-us/azure/postgresql/configure-maintain/concepts-limits

Extension versions by PostgreSQL major
https://learn.microsoft.com/en-us/azure/postgresql/extensions/concepts-extensions-versions

Supported PostgreSQL versions
https://learn.microsoft.com/en-us/azure/postgresql/configure-maintain/concepts-supported-versions

Azure Database for PostgreSQL overview (regions, HA, V6)
https://learn.microsoft.com/en-us/azure/postgresql/overview

Compute
https://learn.microsoft.com/en-us/azure/postgresql/compute-storage/concepts-compute

Storage
https://learn.microsoft.com/en-us/azure/postgresql/compute-storage/concepts-storage

Premium SSD v2
https://learn.microsoft.com/en-us/azure/postgresql/compute-storage/concepts-storage-premium-ssd-v2

Reliability, HA and SLA
https://learn.microsoft.com/en-us/azure/reliability/reliability-database-postgresql

Read replicas
https://learn.microsoft.com/en-us/azure/postgresql/read-replica/concepts-read-replicas

Backup and restore
https://learn.microsoft.com/en-us/azure/postgresql/backup-restore/concepts-backup-restore

Data encryption
https://learn.microsoft.com/en-us/azure/postgresql/security/security-data-encryption

Access control and privileges
https://learn.microsoft.com/en-us/azure/postgresql/security/security-access-control

Private networking
https://learn.microsoft.com/en-us/azure/postgresql/network/concepts-networking-private

Extension considerations
https://learn.microsoft.com/en-us/azure/postgresql/extensions/concepts-extensions-considerations

Major version upgrade
https://learn.microsoft.com/en-us/azure/postgresql/configure-maintain/concepts-major-version-upgrade

Version policy
https://learn.microsoft.com/en-us/azure/postgresql/configure-maintain/concepts-version-policy

Extended support
https://learn.microsoft.com/en-us/azure/postgresql/configure-maintain/extended-support

Release notes
https://learn.microsoft.com/en-us/azure/postgresql/release-notes/release-notes

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